NST DVA · CAPSTONE 2 · SECTION-A · TEAM G12

DeliverIQ

Severity Reality Check

Disentangling true accident severity from congestion, time-of-day, and location bias

ANALYZING 95K US ROAD ACCIDENTS · 2016–2023

Newton School of Technology

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Public severity data is widely used —

The problem: Severity is misleading

WHAT'S WRONG

- Severity measures traffic delay, not crash damage
- Rush-hour minor accidents → Severity 4. Serious low-traffic crashes → Severity 1.
- Insurers, DOTs, and logistics planners use this field directly in pricing models, capital plans, and routing. This introduces systemic bias in decision-making.

OUR CORE BUSINESS QUESTION

How much of what is labeled "severe" in public US accident data is genuinely severe — and how should insurers, DOTs, and logistics planners adjust their decisions to use a bias-corrected signal?

→ *Decision: down-weight raw severity in pricing, capital allocation, and corridor routing.*

Data Engineering

ORIGINAL ROWS	WORKING SAMPLE	COLUMNS	STATES COVERED
7.7M	**95,607**	**49**	**49**
raw Kaggle dataset	*balanced across Sev 1–4*	*incl. 13 derived features*	*5,558 cities*

CLEANING PIPELINE (notebooks/02_cleaning.ipynb)

- Drop high-null + zero-variance columns (End_Lat/Lng, Country, Turning_Loop)
- Validate geographic bounds (lat 24–50, lng −125 to −66)
- Cap Distance_mi outliers at 99th percentile
- Context-aware imputation (median by State + Month + Hour)
- Cast booleans to int for Tableau

KEY DERIVED FEATURES

- Severity_Label · Weather_Category · State_Region
- Time_of_Day · Season · Is_Rush_Hour · Is_Weekend
- Duration_min · Road_Feature_Count
- Is True Severe = Sev=4 AND Distance ≥ 0.5 mi

15 decision-relevant KPIs across 3 dashboards

D1 — Reality Check

the hook

- Total Incidents (95,607)
- Critical Accidents (18,023)
- True Severe Accidents (9,155)
- Avg Duration Severe (1,829 m)
- Severe Distance (24,415 mi)

D2 — When?

the diagnosis

- Worst Season (Winter)
- Peak Risk Hour (12 AM)
- Weekend vs Weekday Δ (+0.34)
- Night Sev 3+ Share (8.42%)
- Riskiest Weather (Drifting Snow)

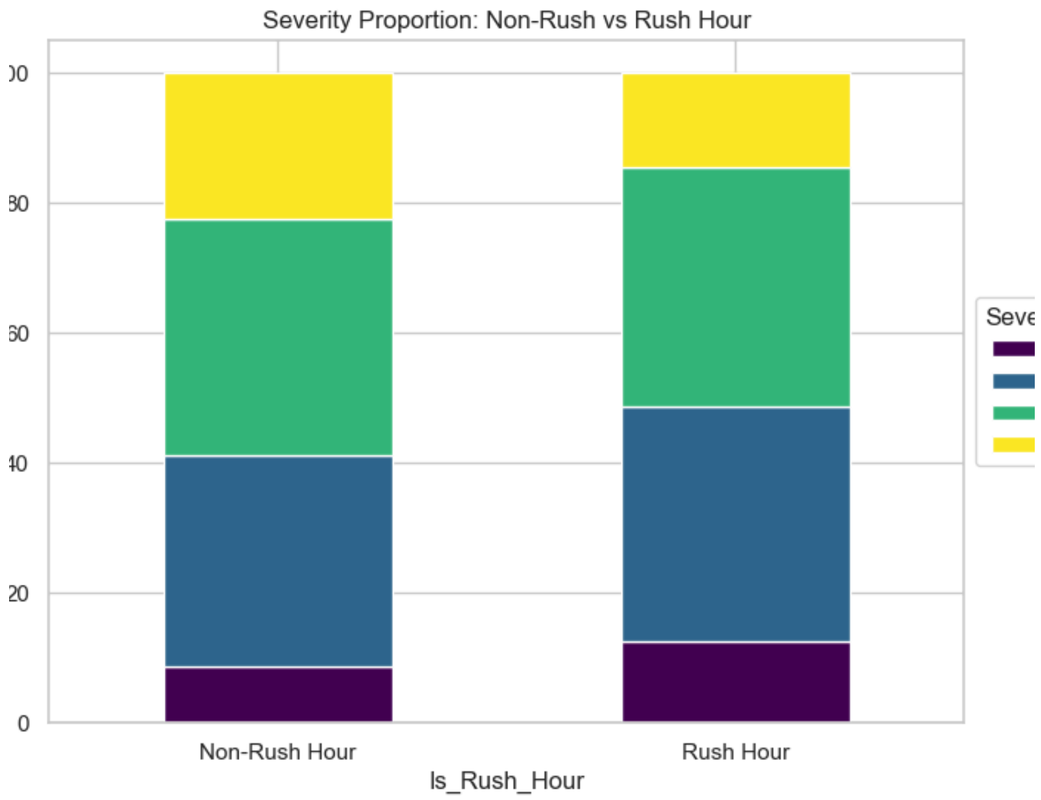
D3 — Where & Why?

the cause

- Top State by Volume (CA · 8,780)
- Top State by Severity (WV)
- Junction Severity (+28.8%)
- Signal Drop (−62.1%)
- Cities Tracked (5,558)

Five observations that reframe the data

- Balanced sampling removes Severity-2 dominance
- Volume peaks at 5 PM rush hour, but per-accident severity is barely different between rush and non-rush.
- Weekday volume rises Tue–Wed, falls weekend. Severity does not follow the same curve.
- Top-volume state: CA (8,780). Top-severity state: WV, geography differs once bias is corrected.
- Most accidents happen in fair weather. Severe accidents are no exception.



Rush vs Non-Rush severity barely changes

Statistical tests aligned with the bias hypothesis

Test 1 — Congestion Bias

Method: Chi-square (Rush Hour × Severity)

Statistically significant, but small practical impact

Test 2 — Duration × Severity

Method: Kruskal-Wallis H (non-parametric)

Median duration rises monotonically: Sev 1 ≈ 30 min → Sev 4 ≈ 210 min. Validates True-Severe definition.

Test 3 — Weather × Severity

Method: Chi-square (top 5 weather × Sev)

Statistically significant but small effect size. Weather shows limited impact on severity.

Test 4 — Location Bias

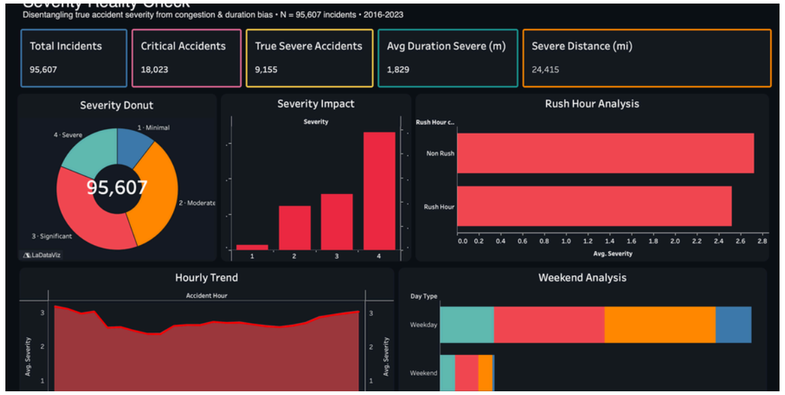
Method: State × Severity cross-tab

WV, PA, MD, NC over-represented in Sev 3+4. Reporting bias is real and must be corrected.

Net finding: Severity in this dataset is partly real, partly noise. The True-Severe filter (Sev 4 + Dist ≥ 0.5 mi) cleanly separates them.

Three dashboards: What → When → Where & Why

D1 Severity Reality Check



True Severe = 9,155 of 18,023 (~51%)

Half of "severe" is congestion noise.

THE HOOK

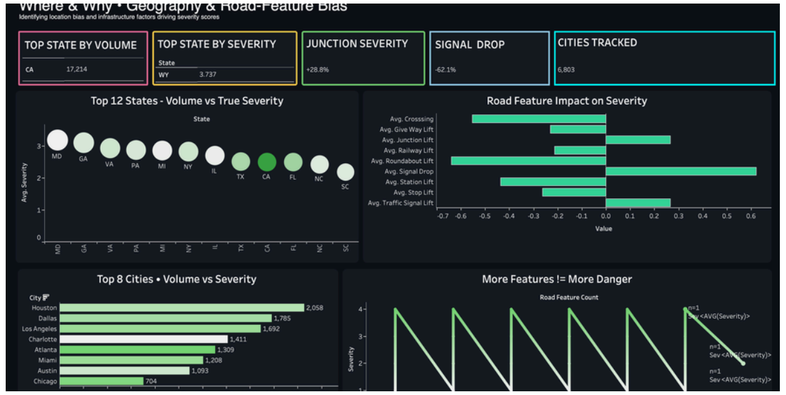
D2 When Risk Strikes

Peak Risk Hour: 12 AM · Worst Season: Winter

Risk peaks at midnight, not rush hour.

THE DIAGNOSIS

D3 Where & Why



Junction +28.8% · Signal –62.1% · Top: WV

Infrastructure has measurable lifts.

THE CAUSE

Interactive filters: Rush Hour · Severity · Visibility Bucket · Time of Day (across 30+ worksheets)

Five actions, each tied to an insight

#	STAKEHOLDER	RECOMMENDATION	EXPECTED IMPACT
1	**Insurer**	Discount publicly-reported Severity 4 by ~50%; switch to True-Severe-based pricing.	*Mis-priced urban-policy margin recovered.*
2	**State DOT**	Prioritize signal installations at top junction hotspots (D3).	*~62% severity reduction at signal-equipped sites; per-installation ROI.*
3	**Logistics fleet**	Re-rank corridors by True-Severe Index; shift dispatch off PA/WV winter night windows.	*Lower expected claim frequency / severity per million miles.*
4	**Federal grants**	Reweight per-state safety allocation using True-Severe counts, not raw volume.	*Capital tracks real severity, not reporting frequency.*
5	**Ride-share / Fleet**	Adjust state-level driver pay & rest-cycle by True-Severe density.	*Driver-safety alignment with actual risk; lower carrier liability.*

Why this work pays off — directional estimates

INSURER · REC #1

~50%

of Severity-4 in pricing models is over-weighted; bias-correction unlocks ~$100M risk-adj premium reallocation per $10B book.

STATE DOT · REC #2

–62%

slower observed severity at locations with traffic signals vs. without — quantifies per-installation infrastructure ROI.

FLEETS · REC #3

5–10%

expected reduction in severe-corridor exposure when routing on True-Severe Index instead of raw volume.

WHY ACT NOW

- Real-time accident APIs increasingly feed pricing engines — bias compounds quickly.
- Post-COVID vehicle-miles-traveled have rebounded; more decisions flow through the same biased input.
- Signal Drop is one of the most cost-effective infrastructure levers — and the data quantifies it.

Honesty about what this analysis can and cannot say

LIMITATIONS

- Severity is a flow-impact label, not a crash-injury label — True Severe is a proxy.
- 0.5-mi threshold is defensible but not absolute; sensitivity tested at 0.25 and 1.0 mi.
- All findings are correlational, not causal.
- API coverage varies by state; 2020+ volume spike is partly artefact.
- No injury / fatality outcomes available in this dataset.
- Balanced sample changes base rates — most KPIs use ratios to control.

NEXT STEPS

- Join FARS / state crash records to validate True-Severe against real injury data.
- Add quasi-experimental causal layer for Signal Drop estimate.
- Wrap corrected severity index as a real-time API for fleets and insurers.
- Train a gradient-boosted classifier on incoming reports (predict True Severe).
- Geospatial drill-down map (Mapbox / Leaflet) for DOT use.
- Attach cost-benefit modelling using FHWA retrofit cost data.

Team G12 · Section A · DeliverIQ

TEAM ROLES

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ARTEFACTS · GitHub + Tableau + Reports

- **GitHub Repo:** github.com/shiavm006/Section-A_G12_DeliverIQ
- **Tableau Public:** public.tableau.com/views/RoadAccidentDataofUSA_17773613139680/Dashboard1
- **Project Report:** reports/project_report.pdf (18 sections, ~22 pages)
- **This Deck:** reports/presentation.pdf (11 slides)

Submitted Apr 29, 2026 · Newton School of Technology · DVA Capstone 2